

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12405294
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Lehmann; Sören et al.

Current sensor

Abstract

A current sensor with an electrical conductor which has a first connection and a second connection for contacting a power circuit of a vehicle, and with a first and a second current measurement device for detecting at least one measurement value for the current flowing over the electrical conductor, and with a first evaluation circuit which is connected to the first and/or the second current measurement device and can receive the measurement values of the first and/or the second current measurement device and output a current measurement signal depending on the measurement values, and with a second evaluation circuit.

Inventors: Lehmann; Sören (Frankfurt am Main, DE), Irsigler; Michael (Regensburg, DE), Gopinath; Anjana (Maruthi Nagar, IN), Weigert; Wolfgang (Tegernheim, DE), Frimberger; Manfred (Ergolsbach, DE)

Applicant: Continental Automotive Technologies GmbH (Hannover, DE)

Family ID: 87280844

Assignee: Continental Automotive Technologies GmbH (Hannover, DE)

Appl. No.: 18/361448

Filed: July 28, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240036083 A1	Feb. 01, 2024

Foreign Application Priority Data

DE	10 2022 207 854.3	Jul. 29, 2022
----	-------------------	---------------

Publication Classification

Int. Cl.: G01R15/20 (20060101); G01R15/14 (20060101); G01R31/382 (20190101)

U.S. Cl.:

CPC G01R15/202 (20130101); G01R15/142 (20130101); G01R31/382 (20190101);

Field of Classification Search

CPC: G01R (31/382); G01R (31/364); G01R (1/203); G01R (35/00)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10184958	12/2018	Ausserlechner	N/A	N/A
10241153	12/2018	Rink	N/A	G01R 1/203
11555832	12/2022	Friedrich	N/A	G01K 7/22
11940470	12/2023	Messier et al.	N/A	N/A
2003/0001559	12/2002	Goto et al.	N/A	N/A
2010/0211347	12/2009	Friedrich	324/202	G01R 33/0035
2012/0262151	12/2011	Mandic et al.	N/A	N/A
2014/0210479	12/2013	Rink et al.	N/A	N/A
2014/0212714	12/2013	Butzmann	N/A	N/A
2014/0333314	12/2013	Kaupp	N/A	N/A
2017/0261536	12/2016	Chae et al.	N/A	N/A
2021/0039576	12/2020	Kaindl	N/A	G01R 31/006
2022/0283200	12/2021	Friedrich et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10060490	12/2001	DE	N/A
102011088893	12/2012	DE	N/A
102014216419	12/2013	DE	G01R 35/005
102014216419	12/2015	DE	N/A
102014219238	12/2015	DE	N/A
1213189	12/2001	EP	N/A
2511714	12/2011	EP	N/A
2732295	12/2013	EP	N/A
2009281773	12/2008	JP	N/A
20160111166	12/2015	KR	N/A

OTHER PUBLICATIONS

Search Report dated Dec. 5, 2023 from corresponding European patent application No. 23184893.8. cited by applicant

Search Report dated May 26, 2023 from related German patent application No. 10 2022 207 853.5. cited by applicant

Search Report dated Dec. 5, 2023 from related European patent application No. 23184892.0. cited by applicant
U.S. Office Action dated Feb. 26, 2025 in the cross referenced U.S. Appl. No. 18/350,948. cited by applicant
European Examination Report dated Feb. 3, 2025 corresponding to European Patent Application No. 23184893.8. cited by applicant
German Search Report dated May 25, 2023 for the German Patent Application No. 10 2022 207 854.3 which this application claims priority. cited by applicant

Primary Examiner: Hawkins; Dominic E

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application claims priority from German Patent Application No. 10 2022 207 854.3 filed on Jul. 29, 2022, in the German Patent and Trade Mark Office, the content of which is herein incorporated by reference.

BACKGROUND

1. Field

(2) Embodiments of the present application relate to a current sensor, in particular for a vehicle battery in a vehicle, with an electrical conductor which has a first connection and a second connection for contacting a power circuit of a vehicle, and with a first current measurement device and a second current measurement device for detecting at least one measurement value for the current flowing over the electrical conductor, and with a first evaluation circuit which is connected to the first current measurement device and/or the second current measurement device and can receive the measurement values of the first current measurement device and/or the second current measurement device and output at least one current measurement signal depending on the measurement values.

2. Description of Related Art

(3) In vehicles with batteries, in particular in electric vehicles or hybrid vehicles, in which batteries are used for the energy supply of the drive unit, it is necessary to permanently monitor the batteries or their state during operation. It is in particular necessary to permanently monitor the charging state of the batteries. Current sensors which are arranged in the current path and detect both the charging currents and the discharging currents of the batteries are used in the vehicles for this purpose.

(4) In order to guarantee the reliability required in the automotive sector and to ensure the required accuracy of the current measurement, it is necessary to provide a redundant current measurement. This redundant current measurement is preferably effected using two independent physical measurement principles in order to avoid or minimize measurement errors due to systematic errors or due to errors which are inherent in one measurement principle. The use of different measurement principles is also referred to as asymmetric redundancy. Such a sensor is shown, for example, in EP2732295A1.

(5) The structure and the production of such sensors are, however, very costly.

SUMMARY

(6) Aspects and objects of embodiments of the present application provide a current sensor with a redundant current measurement which has a high degree of measurement accuracy and a high degree of reliability and is simple to produce.

(7) In order to achieve the object, a current sensor, in particular for a vehicle battery in a vehicle, is provided, with an electrical conductor which has a first connection and a second connection for contacting a power circuit of a vehicle, and with a first current measurement device and a second current measurement device for detecting at least one measurement value for the current flowing over the electrical conductor. The current sensor furthermore has a first evaluation circuit which is connected to the first current measurement device and/or the second current measurement device and can receive the measurement values of the first current measurement device and/or the second current measurement device and output a current measurement signal depending on the measurement values. A second evaluation circuit is furthermore provided, wherein a communications link is provided between the first evaluation circuit and the second evaluation circuit. The first evaluation circuit can send a first test signal to the second evaluation circuit and/or the second evaluation circuit can send a second test signal to the first evaluation circuit. The first evaluation circuit can check the second test signal and/or the second evaluation circuit can check the first test signal. The first evaluation circuit can send a first correction signal in the case of a deviation of the second test signal from a second reference value, and the second evaluation circuit can send a second correction signal in the case of deviation of the first test signal from a first reference value.

(8) The second evaluation circuit can have just the function of a monitoring circuit and not undertake evaluation of the measurement values of the current measurement devices. However, the second evaluation circuit can optionally also receive the measurement values of the current measurement devices and evaluate them.

(9) In principle, the first and the second evaluation circuit have the function of monitoring each other and, in the case of a malfunction of the respective other evaluation circuit, take suitable measures in order to remedy the malfunction of the respective other evaluation circuit. If one of the evaluation circuits identifies a malfunction of the respective other evaluation circuit, a correction signal is sent by virtue of which a suitable measure is taken for the respective other evaluation circuit.

(10) An increased level of safety inside the current sensor is provided by the mutual checking of the evaluation circuits because malfunctions inside the current sensor can be identified and corrected.

(11) The communications link is, for example, a bidirectional link, wherein the test signals of both evaluation circuits are sent via this link. However, the communications link can also have a plurality of lines, for example a separate line for each test signal, in order to be able to identify errors in the communications link.

(12) For example, the first evaluation circuit can have a first correction circuit for receiving the first correction signal and the second evaluation circuit a second correction circuit for receiving the second correction signal, wherein the first correction circuit is provided for triggering a correction procedure for the first evaluation circuit and the second correction circuit is provided for triggering a correction procedure for the second evaluation circuit. The measures for correcting the malfunction are taken, after initiation by the respective other evaluation circuit with the correction signal, by the respective evaluation circuit itself, wherein, by sending the test signal, a check by the respective other evaluation circuit can then take place as to whether the correction of the malfunction was successful.

(13) The correction procedure comprises, for example, a restart of the evaluation circuit and/or a reset of the evaluation circuit. In particular, the first evaluation circuit and/or the second evaluation circuit can have a first or a second correction circuit which receives the correction signal and takes corrective measures, in particular a restart or a reset of the respective evaluation circuit. An additional correction circuit ensures that the triggering of the corrective measures is not prevented by a faulty evaluation circuit itself.

(14) The test signal can contain various pieces of information which enable a diagnosis of the respective other evaluation circuit. Reference values are preferably stored in the respective

evaluation circuit for the information sent with the test signal or they are received or calculated by the respective evaluation circuit itself. In particular, threshold values can be defined for the information by which the information can deviate from the reference values.

(15) In particular, the test signals are sent periodically, in particular at regular time intervals. Consequently, the absence of a test signal can additionally be detected and identified as a malfunction of the respective evaluation circuit. A request signal can optionally also be sent which causes the respective evaluation circuit to send the test signal.

(16) For example, additional information can also be requested via the request signal in order to be able to perform a more accurate diagnosis of the respective other evaluation circuit.

(17) For example, the first test signal contains status information and/or operating information of the first evaluation circuit and/or the second test signal contains status information and/or operating information of the second evaluation circuit. If this status information and/or this operating information deviates from previously defined parameters, it has to be assumed that there is a malfunction of the respective evaluation circuit such that corrective measures are necessary.

(18) The first test signal can also contain measurement values and/or a current measurement signal of the first evaluation circuit and/or the second test signal can contain measurement values and/or a current measurement signal of the second evaluation circuit. By comparing the measurement values and/or the current measurement signals, it is possible to check not only whether the respective evaluation circuit is functioning but also whether it correctly receives and evaluates the measurement values of the respective current measurement device.

(19) The first test signal and/or the second test signal can furthermore contain status information and/or operating information of the first current measurement device and/or the second current measurement device.

(20) In order to improve the reliability of the current sensor or the functional check, the first evaluation circuit and the second evaluation circuit can have separate power supplies and/or separate outlets for a current measurement signal. A separate power supply to the two evaluation circuits can ensure that functional errors due to a faulty voltage supply do not affect both evaluation circuits. A malfunction of one of the two evaluation circuits can consequently be identified more reliably. The separate outlets for the current measurement signal enable the separate output of the current measurement signals. They can consequently be sent, for example, via the communications link or made available separately to a higher-level vehicle control system. In particular, further processing of the current measurement signals can take place in a higher-level vehicle control system, for example additional comparison of the measurement values or the current measurement signals of both evaluation circuits.

(21) In principle, the current sensor can evaluate the measurement values of both current measurement devices and output either a current measurement signal from the measurement values of both current measurement devices or separate current measurement signals for both current measurement devices.

(22) The current sensor can optionally also have a first communications interface for outputting the measurement values of the first current measurement device and a second communications interface of the current measurement signal of the second current measurement device. In this embodiment, the measurement values of the first current measurement device are not evaluated and instead only output such that they can be evaluated, for example by a downstream vehicle control system. The measurement values of the second current measurement device are in contrast evaluated and only a current measurement signal output.

(23) This embodiment can have many advantages. On the one hand, the function of the current sensor, in particular the evaluation circuit, can additionally be checked by the evaluation of the measurement values outside the current sensor.

(24) The measurement values and/or the current measurement signals can each be output in analog or digital form. For example, the current measurement signals of the second current measurement

device are output digitally and the measurement values of the first current measurement device are output in analog form, in particular without any prior processing or evaluation.

(25) Because of the high voltages and currents, it may be necessary to secure parts of the current sensor, in particular of the evaluation circuit, against voltage flashover with respect to the communications interfaces and a downstream vehicle control system, for example by galvanic isolation. In particular, in the case of a second current measurement device with a measuring resistor, the measuring contacts are in direct contact with the electric conductor and with the evaluation circuit such that the evaluation circuit must be protected in particular against voltage flashover to the vehicle control system or other parts of the current sensor. A Hall effect sensor, in contrast, works contactlessly such that there may be no need to protect it against a voltage flashover. Because the first current measurement device in this embodiment has a separate communications interface for outputting the measurement values, an electrical connection to one of the evaluation circuits is not necessary. Insulation or protection from the regions which can come into contact with a high voltage is thus not necessary. The current sensor can consequently be configured with a much simpler structure. It is in particular not necessary to protect the communications interface of the first current measurement device additionally against voltage flashovers.

(26) The first current measurement device can have a Hall effect sensor and the electrical conductor can have a measurement portion, with a constriction with a reduced cross-section, in the region of the first current measurement device, wherein the Hall effect sensor is arranged in this constriction.

(27) A Hall effect sensor is usually used with an iron core which encloses the electrical conductor circumferentially. The Hall effect sensor is arranged in a gap of the iron core. A magnetic field which can be detected by the Hall effect sensor is generated in the iron core by the electric current which flows through the electrical conductor. The arrangement of such an iron core and the positioning of the Hall effect sensor requires a lot of effort, however. According to the embodiment, a so-called difference Hall effect sensor is used instead of a Hall effect sensor with an iron core. A recess in which the Hall effect sensor is positioned is provided in the electrical conductor. The magnetic field generated in this constriction or in this recess by the electric current is strong enough that it can be detected by the Hall effect sensor. An iron core for generating or strengthening the magnetic field can thus be dispensed with. The current sensor can thus have a much more compact design. In addition, the structure is much simpler because only the Hall effect sensor has to be positioned and oriented in the recess. Additional orientation and positioning of the iron core and the Hall effect sensor on this iron core are not necessary.

(28) In particular, the constriction is arranged such that a magnetic field which is as strong as possible and can be detected by the Hall effect sensor is provided inside the constriction. In particular, the geometry of the constriction can be determined by taking into account the geometrical shape of the electrical conductor, the measuring region, and the dimensions of the Hall effect sensor, or a desired strength of the magnetic field to be measured.

(29) The constriction can be formed, for example, by at least one slot which extends from at least one face and/or one edge of the electrical conductor. Such a slot is simple to produce and enables simple positioning of the Hall effect sensor in this slot.

(30) The constriction can be designed so that it is symmetrical with respect to a center axis, running in the current direction, of the electrical conductor, in particular mirror-symmetrical with respect to a plane running through the center axis. Influencing of the current lines or the potential lines which is asymmetrical, in particular mirror-symmetrical, with respect to the center axis consequently takes place such that a magnetic field which is as homogeneous as possible can be generated.

(31) The electrical conductor can, for example, have a plate-shaped design and the constriction is formed by two slots which start from opposite edges of the conductor and extend into the electrical conductor symmetrically with respect to the center axis.

(32) The second current measurement device preferably uses a different physical measurement

principle than the first current measurement device. The second current measurement device can, for example, have a measuring resistor and at least one first measuring contact for detecting a first voltage potential and at least one second measuring contact for detecting a second voltage potential, wherein the first measuring contact and the second measuring contact are arranged offset in the current direction. In particular, the first measuring contact can be arranged upstream from the measuring resistor and the second measuring contact arranged downstream from the measuring resistor in the longitudinal direction of the electrical conductor. The voltage drop over the measuring resistor can be calculated by the measured voltage potentials. From the known electrical resistance of the measuring resistor or a measuring section between the measuring contacts and the measured voltage drop, the current flowing over the measuring resistor or the measuring section, i.e. the electrical conductor, can be calculated using Ohm's law.

(33) The second current measurement device is preferably arranged electrically in series with the first current measurement device. In particular, a symmetrical design of the constriction of the first current measurement device can also be advantageous for the second measurement device because a symmetrical influencing of the current lines or the potential lines is effected thereby.

(34) Both current measurement devices are contacted, for example, with the first evaluation circuit which processes the measurement values of both current measurement devices and generates a common current measurement signal or a current measurement signal for each of the current measurement devices.

(35) Whatever the embodiment, however, the first communications interface and/or the second communications interface can also have protection devices against a voltage flashover.

(36) The first evaluation circuit and/or the second evaluation circuit can furthermore have insulation, in particular galvanic isolation, by means of which the first evaluation circuit and/or the second evaluation circuit is insulated with respect to other regions of the current sensor or other parts of a vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Further advantages and features can be found in the following description in conjunction with the appended drawings, in which:

(2) FIG. 1 shows a first embodiment of a current sensor according to the embodiment, and

(3) FIG. 2 shows a second embodiment of a current sensor according to the embodiment.

DETAILED DESCRIPTION

(4) FIG. 1 shows a battery sensor **10** for a vehicle battery **12** in a vehicle. The battery sensor **10** is arranged in a current circuit **14** in which the vehicle battery **12** and at least one consumer **16** are situated, wherein the battery sensor **10** is arranged such that all the charging currents and discharging currents of the vehicle battery **12** can flow over the battery sensor **10** and thus be measured by the latter. A very accurate estimate of the charging state of the vehicle battery **12** is possible using the measured charging currents and discharging currents.

(5) The battery sensor **10** has an electrical conductor **18** which is arranged in the current circuit **14** and has a first connection **20** and a second connection **22**. The connections **20**, **22** can have, for example, a bore **21**, **23** for a screw connection to the current circuit **14**.

(6) The battery sensor **10** furthermore has a first current measurement device **24** and a second current measurement device **25** which can calculate the current flowing over the electrical conductor **18**.

(7) The first current measurement device **24** has a Hall effect sensor **26** with a first Hall cell **28** and a second Hall cell **30**. The electrical conductor **18** has a constriction **32**, which is formed by two slots **34** extending from opposite edges of the electrical conductor, in the region of the Hall effect

sensor **26**.

- (8) The slots **34** are arranged symmetrically with respect to a center axis **36** extending in the longitudinal direction L of the electrical conductor. The slots **34** are arranged such that in each case one of the Hall cells **28**, **30** of the Hall effect sensor **26** is situated in one of the slots **34** or is arranged in the immediate vicinity of the latter.
- (9) A magnetic field which can be detected by the Hall effect sensor **26** is created around the electrical conductor **18** and in the constriction **32** by the current which flows through the electrical conductor **18**. The magnetic field changes with the current which flows through the electrical conductor **18** such that the current flowing through the electrical conductor **18** can be determined from the strength of the magnetic field and the change in the magnetic field. An iron core is usually arranged around the electrical conductor **18** in order to provide a magnetic field which is strong enough that it can be detected by the Hall effect sensor **26**. In the embodiment shown here, a constriction **32** in which the Hall effect sensor **26** is arranged is used instead of an iron core. The Hall effect sensor **26** is a so-called difference Hall effect sensor by which the magnetic field created by the current can be detected in the constriction **32**. It is consequently possible to dispense with an additional iron core.
- (10) The Hall effect sensor **26** is connected via a communications line **38** to a first evaluation circuit **40**. The evaluation circuit **40** can receive the measurement values of the Hall effect sensor **26** which are received by the Hall effect sensor **26** via the communications line **38**, evaluate them, and generate from them a current measurement signal which is, for example, proportional to the current flowing over the electrical conductor **18**.
- (11) For this purpose, the evaluation circuit **40** has a first analog-to-digital converter **42** and a first microcontroller **44**. The current measurement signals generated by the microcontroller **44** can then be output via a first communications interface **46**, for example to a downstream vehicle control system.
- (12) The second current measurement device **25** has a measuring resistor **48** arranged in the current path. A first measuring contact **50** and a second measuring contact **52**, which can each detect a voltage potential at the electrical conductor **18**, are provided upstream and downstream from the measuring resistor **48** in the longitudinal direction L, i.e. in the current direction. The measuring contacts **50**, **52** are connected to the first evaluation circuit **40** via communications lines **54**, **56**. The evaluation circuit **40** has a second analog-to-digital converter **58** which is connected to the microcontroller **44**.
- (13) The voltage drop over a measuring section defined between the measuring contacts **50**, **52** or over the measuring resistor **48** can be measured from the voltage potentials of the measuring contacts **50**, **52** via the analog-to-digital converter **58** and the microcontroller **44**. From this voltage drop and the known resistance of the measuring section or the measuring resistor **48**, the current flowing over the measuring section or the measuring resistor **48**, i.e. the current flowing over the electrical conductor **18**, can be calculated using Ohm's law.
- (14) The microcontroller **44** can output the current measurement signals, calculated from the measurement values of the first current measurement device **24** and the second current measurement device **25**, individually or jointly to the communications interface **46** such that they can be output to a vehicle control system.
- (15) In addition, further adjustments to the measurement values and/or the current measurement signals, for example a calibration or a temperature compensation, can be made in the microcontroller **44**.
- (16) As can be seen in particular in FIG. 1, a potential separation zone **60** is provided in order to avoid a voltage flashover between the regions of the current sensor **10** which are in contact with the electrical conductor **18** and a vehicle electronic system or a vehicle control system. The potential separation zone has a protection device **62** in order to be able to transmit the current measurement signals to a vehicle electronic system or a vehicle control system in the case of sufficiently high

protection against a voltage flashover. The communications interface **46** can, for example, also be integrated into the protection device **62**.

(17) The first evaluation circuit **40** furthermore has a correction circuit **64** via which suitable measures can be initiated in the case of a malfunction of the evaluation circuit **40** in order to remedy the malfunction of the evaluation circuit **40**. For example, the correction circuit **64** can effect a restart or a reset of the evaluation circuit **40**.

(18) A second evaluation circuit **66** is furthermore provided which is connected to the first evaluation circuit **40** via a communications link **68**. The communications link **68** is, for example, a bidirectional communications line via which information can be exchanged between the first evaluation circuit **40** and the second evaluation circuit **66**.

(19) The second evaluation circuit **66** furthermore has a second correction circuit **70** and at least one inlet **72** for operating information of the current sensor **10**, for example of the microcontroller **44**, of the evaluation circuit **40**, of the first current measurement device **24**, and/or of the second current measurement device **25**. The inlet **72** is connected to an analog-to-digital converter **74**.

(20) The first evaluation circuit **40** and the second evaluation circuit **66** can exchange information via the communications link **68** in order to check the proper operating state of the respective other evaluation circuit **66**, **40**. Each evaluation circuit **40**, **66** preferably sends via the communications link **68** a test signal with previously defined information which is checked by the respective other evaluation circuit **66**, **40**. If the information contained in the test signal does not match the information which is stored in the evaluation circuit **66**, **40** or is received or calculated by the latter, a correction signal is sent to the correction circuit **64**, **70** of the respective other evaluation circuit **40**, **66**. For example, the respective evaluation circuit **40**, **66** is restarted or reset on the basis of this correction signal.

(21) The test signals can, for example, contain operating information or status information of the first current measurement device **24**, of the second current measurement device **25**, and/or of the respective other evaluation circuit **40**, **66**.

(22) If operating information or status information of the first current measurement device **24** and/or of the second current measurement device **25** is contained in the test signal, it can be received and processed or evaluated by both evaluation circuits **40**, **66**. The processed or evaluated operating information or status information can then be transmitted with the test signal via the communications link to the respective other evaluation circuit **40**, **66**. The latter can compare the information contained in the test signal with the information evaluated by the independent evaluation circuit **40**, **66**. If the evaluated information differs from each other, it has to be assumed that the evaluation in the evaluation circuit **40**, **66** is incorrect and hence the evaluation circuit **40**, **66** is not working correctly. If this is the case, a correction signal can then be generated by the respective other evaluation circuit **66**, **40** and sent to the evaluation circuit **40**, **66**.

(23) The operating information or status information is, for example, operating voltages of the first current measurement device **24**, of the second current measurement device **25**, and/or of the evaluation circuit **40**, **66**. The operating voltages of the current measurement devices **24**, **25** are, for example, provided by the first evaluation circuit **40** such that a check can also be made using the above described method as to whether the correct operating voltages are provided. In particular, in this embodiment the operating voltages of the current measurement device **24**, **25** can be sent directly via an outlet **74** to the second evaluation circuit **66** or the inlet **72** of the second evaluation circuit **66** and checked by the latter.

(24) Alternatively, the test signals can also contain measurement values and/or information calculated from the measurement values, for example a current measurement signal. The measurement values received by the two evaluation circuits **40**, **66** and/or the evaluated information can be compared with each other. If they deviate, it has to be assumed that there is a malfunction of the evaluation circuit **40**, **66**.

(25) The test signals are, for example, sent periodically at previously defined time intervals. The

absence of a test signal or delayed sending of a test signal can consequently also be evaluated as a malfunction of the respective evaluation circuit **40, 66**.

(26) In particular, after sending the correction signal and a subsequent restart or reset of the evaluation circuit **40, 66**, a new test signal can also be directly sent in order to check whether the restart or the reset of the evaluation circuit **40, 66** was successful and the malfunction has been remedied. In such a case, a test signal can, for example, also be requested from the evaluation circuit **40, 66** via the communications link **68**.

(27) The embodiment shown in FIG. 2 likewise has an electrical conductor **18** with a first current measurement device **24** and a second current measurement device **25**. The functioning of the current measurement devices **24, 25** corresponds to the embodiment shown in FIG. 1. The first current measurement device **24** has a Hall effect sensor **26**. The second current measurement device **25** has a measuring resistor **48** and two measuring contacts **50, 52**. A first evaluation circuit **40** is furthermore provided which is connected to the measuring contacts **50, 52** of the second current measurement device **25**. The structure of the first evaluation circuit **40** corresponds substantially to the first evaluation circuit **40** shown in FIG. 1.

(28) In contrast to the embodiment shown in FIG. 1, however, the Hall effect sensor **26** has a separate voltage supply **76** and separate second communications interfaces **78, 80** for outputting the measurement values and/or an error signal, operating information, and/or status information.

(29) The communications interface **78** can output the measurement values of the Hall effect sensor **26** unchanged, for example to a vehicle control system. The measurement values of the Hall effect sensor **26** can be evaluated or processed in the vehicle control system. The evaluated measurement values of the Hall effect sensor **26** can then be compared with the current measurement signals, output via the first communications interface **46**, of the second current measurement device **25**.

(30) As can be seen in FIG. 2, the communications interfaces **78, 80** and the voltage supply **76** have protection devices **82, 84, 86** against a positive and/or negative overvoltage.

(31) Furthermore, just the first evaluation circuit **40** is provided in the potential separation zone **60**. Because the measuring contacts **50, 52** are in direct contact with the electrical conductor **18**, it is also necessary in this embodiment to separate or insulate them relative to the other regions of the current sensor and relative to a vehicle control system. Because the Hall effect sensor **26** measures contactlessly and thus is not in conductive contact with the electrical conductor **18**, it can be arranged outside the potential separation zone **60**. Additional protection against a positive and/or negative overvoltage is nevertheless provided by the protective devices **82, 84, 86**.

(32) Furthermore, no communications link **68** is provided in the embodiment shown here. A comparison of the measurement values, the current measurement signals, the operating information, and/or the status information can be made in the vehicle control system in this embodiment. For this reason, information from which a test signal can be calculated, for example error signals, operating information, or status information of the Hall effect sensor **26** is, for example, output via the communications interface **80**. The information of the first evaluation circuit **40** can furthermore be output unprocessed or as a test signal via the first communications interface **46**.

Claims

1. A current sensor comprising: an electrical conductor comprising a first connection and a second connection configured to contact a battery; a first current measurement device configured to detect a first measurement value of a current flowing over the electrical conductor; a second current measurement device configured to detect a second measurement value of the current flowing over the electrical conductor; a first evaluation circuit connected to the first current measurement device, the first evaluation circuit configured to receive the first measurement value and output a first current measurement signal based on the first measurement value; a second evaluation circuit

connected to the second current measuring device, the second evaluation circuit configured to receive the second measurement value and output a second current measurement signal based on the second measurement value; and a communications link provided between the first evaluation circuit and the second evaluation circuit, wherein the first evaluation circuit is configured to send a first test signal to the second evaluation circuit, wherein the second evaluation circuit is configured to send a second test signal to the first evaluation circuit, wherein the first evaluation circuit is configured to check the second test signal and send a first correction signal to the second evaluation unit when the second test signal deviates from a second reference value, wherein the second evaluation circuit is configured to check the first test signal and send a second correction signal to the first evaluation unit when the first test signal deviates from a first reference value.

2. The current sensor as claimed in claim 1, wherein the first evaluation circuit comprises a first correction circuit configured to receive the first correction signal, wherein the second evaluation circuit comprises a second correction circuit configured to receive the second correction signal, wherein the first correction circuit is configured to trigger a first correction procedure for the first evaluation circuit, and wherein the second correction circuit is configured to trigger a second correction procedure for the second evaluation circuit.

3. The current sensor as claimed in claim 2, wherein the first correction procedure comprises a restart of the first evaluation circuit, and wherein the second correction procedure comprises a restart of the second evaluation circuit.

4. The current sensor as claimed in claim 3, wherein the first test signal contains status information and/or operating information of the first evaluation circuit, and wherein the second test signal contains status information and/or operating information of the second evaluation circuit.

5. The current sensor as claimed in claim 4, wherein the first test signal contains measurement values and/or a current measurement signal of the first evaluation circuit, and wherein the second test signal contains measurement values and/or a current measurement signal of the second evaluation circuit.

6. The current sensor as claimed in claim 5, wherein the first test signal and/or the second test signal contains status information and/or operating information of the first current measurement device and/or the second current measurement device.

7. The current sensor as claimed in claim 6, further comprising: a first power supply coupled to the first evaluation circuit; and a second power supply coupled to the second evaluation circuit.

8. The current sensor as claimed in claim 7, wherein the current sensor comprises: a first communications interface configured to output the first current measurement signal; and a second communications interface configured to output the second current measurement signal.

9. The current sensor as claimed in claim 8, further comprising: a first protection device configured to protect the first communications interface against a positive and/or negative overvoltage; and a second protection device configured to protect the second communications interface against a positive and/or negative overvoltage.

10. The current sensor as claimed in claim 9, wherein the first evaluation circuit comprises first galvanic insulation, and wherein the second evaluation circuit comprises second galvanic isolation.

11. The current sensor as claimed in claim 10, wherein the first current measurement device is a Hall effect sensor and the electrical conductor comprises a measurement portion, with a constriction with a reduced cross-section, in the region of the first current measurement device, wherein the Hall effect sensor is arranged in the constriction.

12. The current sensor as claimed in claim 11, wherein the constriction is formed by at least one slot which extends from at least one face and/or one edge of the electrical conductor.

13. The current sensor as claimed in claim 12, wherein the constriction is designed as symmetrical with respect to a center axis, running in the longitudinal direction of the electrical conductor with respect to a plane running through the center axis.

14. The current sensor as claimed in claim 13, wherein the second current measurement device

comprises a measuring resistor and at least one first measuring contact configured to detect a first voltage potential and at least one second measuring contact configured to detect a second voltage potential, and wherein the first measuring contact and the second measuring contact are arranged offset in the longitudinal direction.
